

Homework 1 warm-up



hw0.pdf
PDF



hw1 (2).pdf
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Make me a "warm-up" for homework 1. I've included hw 0, which is pre-requisites, and the discussion worksheet. Make the worksheet include all necessary content, and questions for me to answer before doing hw 1.

EECS 182 – Homework 1 Warm-Up Worksheet

(Based on HW0 and Discussion 1 – do this *before* HW1)

This worksheet is just for you; not for submission. It's meant to refresh exactly the math and ideas HW1 leans on from:

- HW0 Problems 2–4 (vector calculus, least squares, ridge) 
- HW0 Problem 3–4 (SVD, min-norm, ridge interpretations) 
- Discussion 1, "Gradient Descent Mechanics" (constant + linear models, SGD) 
- HW1 Problems 1–7 (what you're about to do) 

Try to be able to do this whole sheet without notes; if you get stuck somewhere, that's a red flag for that HW1 problem.

1. Vector & Matrix Calculus Refresher (HW0 Problem 2 vibes)

Let $x, c \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$. We follow the convention that gradients of scalars wrt column vectors are row vectors.

1.1 For $f(x) = x^T c$:

- (a) What is the shape of $\partial f / \partial x$ (scalar, vector, or matrix)?
- (b) Compute $\partial f / \partial x$ explicitly, as a row vector in terms of c .

1.2 For $f(x) = \|x\|_2^2 = x^T x$:

- (a) Rewrite $f(x) = \sum_{i=1}^n x_i^2$.
- (b) Compute $\partial f / \partial x$ as a row vector; simplify in compact matrix notation.

1.3 For $f(x) = Ax$ (a vector-valued function):

- (a) What is the shape of $\partial f / \partial x$ (Jacobian)?
- (b) Write the Jacobian explicitly in terms of A .

1.4 For $f(x) = x^T A x$ (a scalar):

- (a) Expand $f(x)$ as a double sum $\sum_i \sum_j a_{ij} x_i x_j$.
- (b) Compute $\partial f / \partial x_i$ for a fixed i .
- (c) Stack your answers into $\partial f / \partial x$ and write it cleanly in matrix form.
- (d) Under what condition on A does this derivative simplify to $2 x^T A$?

2. SVD, Least Squares, and Min-Norm Solutions (HW0 Problem 3)

Let $X \in \mathbb{R}^{m \times n}$, $y \in \mathbb{R}^m$, and full SVD $X = U \Sigma V^T$ with $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$, $\Sigma \in \mathbb{R}^{m \times n}$.

2.1 Overdetermined case ($m > n$, full column rank):

We solve $\min_w \|Xw - y\|_2^2$.

- (a) Using normal equations, derive w^* in terms of X and y .
- (b) Substitute $X = U \Sigma V^T$ and simplify w^* as $V \Sigma^\dagger U^T y$ where $\Sigma^\dagger$ is the pseudoinverse diagonal matrix with entries $1/\sigma_i$. Explicitly show the algebraic steps.

2.2 Underdetermined case ($m < n$, full row rank):

We want the minimum-norm solution satisfying $Xw = y$.

- (a) Write the optimization problem that defines the min-norm solution.
- (b) Derive the solution $w_{\min}$ in terms of X and y using Lagrange multipliers or otherwise.
- (c) Express $w_{\min}$ in SVD form (using U, Σ, V); show that it equals $V \Sigma^\dagger U^T y$ as well.

2.3 Concept check:

- (a) In one sentence each, explain the meaning of “least-squares solution” vs “minimum-norm solution”.
- (b) In which case ($m > n$ or $m < n$) are these unique?

3. Ridge Regression Basics (HW0 Problem 4 seed)

Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, $\lambda > 0$. Consider ridge regression:

$$L(w) = \|y - Xw\|_2^2 + \lambda \|w\|_2^2.$$

3.1 Closed-form solution:

- (a) Take the gradient of $L(w)$ wrt w and set it to zero.
- (b) Solve for w_{ridge} in terms of X, y, λ .

3.2 SVD perspective and shrinkage:

Let $X = U \Sigma V^T$ as before.

- (a) Substitute $X = U \Sigma V^T$ into your formula for w_{ridge} and simplify until the only non-identity diagonal matrix is of the form $\text{diag}(\text{some scalar function of } \sigma_i)$.
- (b) What scalar factor multiplies each component of $U^T y$ in the V basis? Write it as a function of σ_i and λ .

- (c) Describe what happens to directions with very small σ_i and very large σ_i as λ grows.

3.3 Interpretation:

- (a) Explain in your own words why ridge regression is called “shrinkage”.
- (b) As $\lambda \rightarrow 0$, what does w_{ridge} tend to (assuming $X^T X$ is invertible)?
- (c) As $\lambda \rightarrow \infty$, what vector does w_{ridge} approach?

4. Scalar Gradient Descent Stability (prep for HW1 Problem 1)

Consider the scalar objective:

$$L(w) = (y - \sigma w)^2, \text{ with } \sigma > 0.$$

We run gradient descent with learning rate η :

$$w_{t+1} = w_t - \eta \frac{dL}{dw} \text{ (evaluated at } w_t \text{)}.$$

4.1 Derive the update:

- (a) Compute dL/dw .
- (b) Write the explicit recurrence w_{t+1} in terms of w_t , y , σ , η .
- (c) Identify the optimal solution w^* that minimizes L .

4.2 Error dynamics:

$$\text{Let } e_t = w_t - w^*.$$

- (a) Derive a recurrence relation for e_{t+1} in terms of e_t .
- (b) Give the common ratio $r(\eta, \sigma)$ such that $e_t = r^t e_0$. What is r ?

4.3 Stability region:

- (a) For gradient descent to converge, what condition must $|r|$ satisfy?
- (b) Use that condition to find the range of $\eta > 0$ for which the recurrence is stable. Express it as an inequality involving σ .

4.4 Convergence speed:

- (a) Suppose $|e_t| \leq \alpha^t |e_0|$ for some $\alpha \in (0,1)$. How many steps t are needed to make $|e_t| \leq \epsilon |e_0|$?
- (b) Plug in $\alpha = |1 - 2\eta\sigma^2|$ and express t in terms of η , σ , and a target factor $(1 - \epsilon)$.

5. Two-Dimensional Diagonal System and Condition Number (HW1 Problem 1 extension)

Now consider a 2D system:

$$\begin{bmatrix} \sigma & 0 \\ 0 & \sigma_s \end{bmatrix} \begin{bmatrix} w[1] \\ w[2] \end{bmatrix} = \begin{bmatrix} y[1] \\ y[2] \end{bmatrix}$$

with $\sigma_\ell \gg \sigma_s > 0$. We run gradient descent with a *single* learning rate η on the squared loss.

5.1 Mode-wise behavior:

- (a) Argue that each coordinate $w[1]$, $w[2]$ behaves like an independent scalar gradient descent on a quadratic with “ σ ” equal to σ_ℓ and σ_s , respectively.
- (b) Write the error recurrence $e_t^{(1)}$ and $e_t^{(2)}$ and the corresponding ratios $r_\ell(\eta)$ and $r_s(\eta)$.

5.2 Stability and limiting factor:

- (a) What η range must hold to keep *both* dimensions stable?
- (b) Which σ (σ_ℓ or σ_s) limits how large η can be while still converging?

5.3 Relative convergence speeds:

- (a) For a fixed η in the stable range, which component (associated with σ_ℓ or σ_s) converges faster?
- (b) Define a condition number $\kappa = (\sigma_\ell^2)/(\sigma_s^2)$. In one sentence, describe how κ affects the overall convergence rate.

6. Gradient Descent & SGD on Simple Models (Discussion 1 refresh)

You have N data points (x_i, y_i) .

6.1 Constant model $\hat{y}_i = \theta$

We want to minimize:

$$L(\theta) = (1/N) \sum_{i=1}^N (y_i - \theta)^2.$$

- (a) Compute $\partial L / \partial \theta$.
- (b) Solve for the optimal θ^* in closed form.
- (c) Write the full-batch gradient descent update $\theta \leftarrow \theta - \alpha \partial L / \partial \theta$.
- (d) Write the SGD update when you sample a single i uniformly from $\{1, \dots, N\}$.
- (e) In one sentence, explain why each point has the same “vote” in the gradient.

6.2 Linear model $\hat{y}_i = \theta x_i$

$$\text{Now } L(\theta) = (1/N) \sum_{i=1}^N (y_i - \theta x_i)^2.$$

- (a) Compute $\partial L / \partial \theta$.
- (b) Solve for θ^* in closed form.
- (c) Write the full-batch gradient descent update for θ .
- (d) Write the single-sample SGD update.
- (e) Do all data points contribute equally in magnitude to the full gradient? Explain in terms of x_i .

7. Noise as Regularization (prep for HW1 Problem 5)

For each data point $x_i \in \mathbb{R}^n$ we see noisy versions:

$$\tilde{x}_i = x_i + N_i, \text{ with } N_i \sim \mathcal{N}(0, \sigma^2 I_n) \text{ i.i.d.}$$

We consider minimizing the expected least-squares loss:

$$\min_w E[\|\tilde{X}w - y\|_2^2].$$

7.1 One-dimensional sanity check:

Let $x, y, w \in \mathbb{R}$, $\tilde{x} = x + N$, $N \sim \mathcal{N}(0, \sigma^2)$.

- (a) Compute $E[(\tilde{x}w - y)^2]$ explicitly, expanding and using $E[N] = 0$, $E[N^2] = \sigma^2$.
- (b) Show that this can be written as $(xw - y)^2 + (\text{something}) \cdot w^2$. Identify that “something” in terms of x and σ^2 .

7.2 Matrix case:

Let X be the $m \times n$ matrix of true inputs. Let $\tilde{X}$ be the random noisy matrix with i -th row $\tilde{x}_i^T$.

- (a) Write $\|\tilde{X}w - y\|_2^2$ as an inner product $(\tilde{X}w - y)^T (\tilde{X}w - y)$.
- (b) Take expectation and split into $E[\tilde{X}^T \tilde{X}]$ terms and cross terms. Use that $E[N_i] = 0$, $\text{cov}(N_i) = \sigma^2 I$.
- (c) Show that $E[\|\tilde{X}w - y\|_2^2]$ can be written as $\|Xw - y\|_2^2 + \lambda \|w\|_2^2$ for some λ . Identify λ in terms of σ^2 and possibly X .

7.3 Concept:

- (a) In words: why does adding isotropic Gaussian noise to inputs act like an L2 regularizer?
- (b) If σ^2 increases, what happens to the effective λ ?

8. General Tikhonov / Weighted Least Squares (prep for HW1 Problem 6)

Consider the optimization problem:

$$\min_x \|W_1(Ax - b)\|_2^2 + \|W_2(x - c)\|_2^2,$$

where W_1, W_2, A are matrices and x, b, c are vectors.

8.1 1D toy version:

Let $a, b, c, \alpha > 0$ be scalars, and solve:

$$\min_x (ax - b)^2 + \alpha (x - c)^2.$$

- (a) Expand the expression and take derivative w.r.t. x .
- (b) Set derivative to zero and solve for x^* .
- (c) Interpret x^* as a weighted average of b/a and c . What are the weights?

8.2 Matrix expansion:

Back to the matrix problem:

- (a) Write the objective as $(A x - b)^T W_1^T W_1 (A x - b) + (x - c)^T W_2^T W_2 (x - c)$.
- (b) Take gradient wrt x , set to zero, and solve for x^* in terms of A , b , c , W_1 , W_2 .

8.3 Stacking trick:

- (a) Construct a matrix C and vector d such that $\|W_1 (A x - b)\|_2^2 + \|W_2 (x - c)\|_2^2 = \|C x - d\|_2^2$.
Explicitly write C and d as a vertical concatenation.
- (b) Using the OLS solution $x^* = (C^T C)^{-1} C^T d$, verify it matches the expression from 8.2.
- (c) Choose W_1 , W_2 , and c so that this reduces exactly to standard ridge regression:
 $\min_x \|A x - b\|_2^2 + \lambda \|x\|_2^2$.

9. Gaussian MAP & Ridge Regression (prep for HW1 Problem 7)

Let $W \in \mathbb{R}^d$, $Y \in \mathbb{R}^n$ be random vectors with:

- Prior: $W \sim N(0, I_d)$.
- Noise: $N \sim N(0, I_n)$, independent of W .
- Observation model: $Y = X W + \sqrt{\lambda} N$.

9.1 Covariances:

- (a) Compute $\Sigma_{WW} = E[W W^T]$.
- (b) Compute $\Sigma_{YY} = E[Y Y^T]$ (expand $Y = X W + \sqrt{\lambda} N$ and use independence).
- (c) Compute $\Sigma_{WY} = E[W Y^T]$.

9.2 Conditional expectation formula:

For jointly zero-mean Gaussian (W, Y) :

$$E[W \mid Y = y] = \Sigma_{WY} \Sigma_{YY}^{-1} y.$$

- (a) Plug your matrices from 9.1 into this formula and simplify $E[W \mid Y = y]$ into an expression involving X , λ , and y only.
- (b) Show that this expression matches one of the standard ridge forms:

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y$$

or

$$\hat{w} = X^T (X X^T + \lambda I)^{-1} y.$$

Indicate clearly which one you get and how to see it equals the other.

9.3 Interpretation:

- (a) Explain why the MAP estimate of W given $Y = y$ is the same as the ridge regression solution.
- (b) In words: how does λ control how much you “trust” the prior vs the data?

If you can comfortably get through this warm-up, you should be in very good shape conceptually and technically for all parts of Homework 1.

Sources

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-  **Warm-up worksheet (questions only):**
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yes do latex of both

Here are clean LaTeX sources for both documents.

1 Worksheet LaTeX (eecs182_hw1_warmup.tex)

latex

```
\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
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\usepackage{amsmath,amssymb,amsfonts}
\usepackage{bm}

\title{EECS 182 -- Homework 1 Warm-Up Worksheet}
\author{}
\date{}

\begin{document}
\maketitle

\noindent
\textbf{Instructions.} This worksheet is just for you; not for submission. It is meant to
refresh exactly the math and ideas Homework 1 leans on from HW0 and Discussion 1. Try to be able
to do this whole sheet without notes; if you get stuck, that flags a topic to review.

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\section{Vector and Matrix Calculus Refresher}

Let  $\mathbf{x}, \mathbf{c} \in \mathbb{R}^n$  and  $A \in \mathbb{R}^{n \times n}$ . We follow the
convention that gradients of scalars with respect to column vectors are row vectors.

\subsection*{1.1  $f(\mathbf{x}) = \mathbf{x}^{\top} \mathbf{c}$ }
\begin{enumerate}
\item(a) What is the shape of  $\frac{\partial f}{\partial \mathbf{x}}$  (scalar, vector, or
matrix)?
\item(b) Compute  $\frac{\partial f}{\partial \mathbf{x}}$  explicitly as a row vector in terms
of  $\mathbf{c}$ .
\end{enumerate}

\subsection*{1.2  $f(\mathbf{x}) = \|\mathbf{x}\|_2^2 = \mathbf{x}^{\top} \mathbf{x}$ }
\begin{enumerate}
\item(a) Rewrite  $f(\mathbf{x})$  as  $\sum_{i=1}^n x_i^2$ .
\item(b) Compute  $\frac{\partial f}{\partial \mathbf{x}}$  as a row vector and simplify it in
compact matrix notation.
\end{enumerate}

\subsection*{1.3  $f(\mathbf{x}) = A \mathbf{x}$  (vector-valued)}
\begin{enumerate}
\item(a) What is the shape of  $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}$  (the Jacobian)?
\item(b) Write the Jacobian explicitly in terms of  $A$ .
\end{enumerate}

\subsection*{1.4  $f(\mathbf{x}) = \mathbf{x}^{\top} A \mathbf{x}$ }
\begin{enumerate}
\item(a) Expand  $f(\mathbf{x})$  as a double sum  $\sum_i \sum_j a_{ij} x_i x_j$ .
\item(b) Compute the partial derivative  $\frac{\partial f}{\partial x_i}$  for a fixed
 $i$ .
\item(c) Stack your answers into  $\frac{\partial f}{\partial \mathbf{x}}$  and write it
cleanly in matrix form.
\item(d) Under what condition on  $A$  does this derivative simplify to  $2 \mathbf{x}^{\top} A$ ?
\end{enumerate}

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\section{SVD, Least Squares, and Minimum-Norm Solutions}

Let  $X \in \mathbb{R}^{m \times n}$ ,  $\mathbf{y} \in \mathbb{R}^m$ , and take the full SVD

$$X = U \Sigma V^{\top}, \quad U \in \mathbb{R}^{m \times m}, \quad V \in \mathbb{R}^{n \times n}, \quad \Sigma \in \mathbb{R}^{m \times n}.$$


\subsection*{2.1 Overdetermined case ( $m > n$ , full column rank)}
We solve

$$\min_{\mathbf{w}} \|\mathbf{X} \mathbf{w} - \mathbf{y}\|_2^2.$$

\begin{enumerate}
\item(a) Using the normal equations, derive  $\mathbf{w}^*$  in terms of  $X$  and  $\mathbf{y}$ .
\item(b) Substitute  $X = U \Sigma V^{\top}$  and simplify  $\mathbf{w}^*$  to the form

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\begin{equation}
\| \mathbf{w} \|^2 = \mathbf{w}^T \mathbf{\Sigma}^+ \mathbf{U}^{\text{top}} \mathbf{y},
\end{equation}
where  $\mathbf{\Sigma}^+$  is the pseudoinverse diagonal matrix with entries  $1/\sigma_i$ . Show the algebraic steps.
\end{enumerate}

\subsection*{2.2 Underdetermined case ( $m < n$ , full row rank)}
We want the minimum-norm solution satisfying  $X \mathbf{w} = \mathbf{y}$ .
\begin{enumerate}
\item[(a)] Write the optimization problem that defines the minimum-norm solution.
\item[(b)] Derive the solution  $\mathbf{w}_{\min}$  in terms of  $X$  and  $\mathbf{y}$  using Lagrange multipliers or otherwise.
\item[(c)] Express  $\mathbf{w}_{\min}$  in SVD form using  $U$ ,  $\mathbf{\Sigma}$ ,  $V$  and show that it equals  $V \mathbf{\Sigma}^+ U^{\text{top}} \mathbf{y}$  as well.
\end{enumerate}

\subsection*{2.3 Concept check}
\begin{enumerate}
\item[(a)] In one sentence each, explain the meaning of "least-squares solution" versus "minimum-norm solution".
\item[(b)] In which case ( $m > n$  or  $m < n$ ) are these solutions unique?
\end{enumerate}

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\section{Ridge Regression Basics}

Let  $X \in \mathbb{R}^{n \times d}$ ,  $\mathbf{y} \in \mathbb{R}^n$ , and  $\lambda > 0$ . Consider ridge regression
\begin{equation}
L(\mathbf{w}) = \| \mathbf{y} - X \mathbf{w} \|^2 + \lambda \| \mathbf{w} \|^2.
\end{equation}

\subsection*{3.1 Closed-form solution}
\begin{enumerate}
\item[(a)] Take the gradient of  $L(\mathbf{w})$  with respect to  $\mathbf{w}$  and set it to zero.
\item[(b)] Solve for  $\mathbf{w}_{\text{ridge}}$  in terms of  $X$ ,  $\mathbf{y}$ , and  $\lambda$ .
\end{enumerate}

\subsection*{3.2 SVD perspective and shrinkage}

Let  $X = U \mathbf{\Sigma} V^{\text{top}}$  as before.
\begin{enumerate}
\item[(a)] Substitute  $X = U \mathbf{\Sigma} V^{\text{top}}$  into your formula for  $\mathbf{w}_{\text{ridge}}$  and simplify until the only non-identity diagonal matrix is of the form  $\text{diag}(\text{some scalar function of } \sigma_i)$ .
\item[(b)] What scalar factor multiplies each component of  $U^{\text{top}} \mathbf{y}$  in the  $V$  basis? Write it as a function of  $\sigma_i$  and  $\lambda$ .
\item[(c)] Describe what happens to directions with very small  $\sigma_i$  and with very large  $\sigma_i$  as  $\lambda$  grows.
\end{enumerate}

\subsection*{3.3 Interpretation}
\begin{enumerate}
\item[(a)] Explain in your own words why ridge regression is called "shrinkage".
\item[(b)] As  $\lambda \rightarrow 0$ , what does  $\mathbf{w}_{\text{ridge}}$  tend to (assuming  $X^{\text{top}} X$  is invertible)?
\item[(c)] As  $\lambda \rightarrow \infty$ , what vector does  $\mathbf{w}_{\text{ridge}}$  approach?
\end{enumerate}

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\section{Scalar Gradient Descent Stability}

Consider the scalar objective
\begin{equation}
L(w) = (y - \sigma w)^2, \quad \sigma > 0.
\end{equation}
We run gradient descent with learning rate  $\eta$ :
\begin{equation}

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w_{t+1} = w_t - \eta \frac{dL}{dw} \Big|_{w_t}.
\]

\subsection*{4.1 Derive the update}
\begin{enumerate}
  \item[(a)] Compute  $\frac{dL}{dw}$ .
  \item[(b)] Write the explicit recurrence  $w_{t+1}$  in terms of  $w_t$ ,  $y$ ,  $\sigma$ , and  $\eta$ .
  \item[(c)] Identify the optimal solution  $w^*$  that minimizes  $L$ .
\end{enumerate}

\subsection*{4.2 Error dynamics}
Let  $e_t = w_t - w^*$ .
\begin{enumerate}
  \item[(a)] Derive a recurrence relation for  $e_{t+1}$  in terms of  $e_t$ .
  \item[(b)] Find the common ratio  $r(\eta, \sigma)$  such that  $e_t = r^t e_0$ . What is  $r$ ?
\end{enumerate}

\subsection*{4.3 Stability region}
\begin{enumerate}
  \item[(a)] For gradient descent to converge, what condition must  $|r|$  satisfy?
  \item[(b)] Use that condition to find the range of  $\eta > 0$  for which the recurrence is stable.
  Express it as an inequality involving  $\sigma$ .
\end{enumerate}

\subsection*{4.4 Convergence speed}
\begin{enumerate}
  \item[(a)] Suppose  $|e_t| \leq \alpha^t |e_0|$  for some  $\alpha \in (0,1)$ . How many steps  $t$  are needed to make  $|e_t| \leq \epsilon |e_0|$ ?
  \item[(b)] Plug in  $\alpha = |1 - 2\eta\sigma^2|$  and express  $t$  in terms of  $\eta$ ,  $\sigma$ , and  $\epsilon$ .
\end{enumerate}

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\section{Two-Dimensional Diagonal System and Condition Number}

Now consider a 2D system
\[\begin{matrix} \begin{bmatrix} \sigma_L & 0 \\ 0 & \sigma_S \end{bmatrix} \begin{bmatrix} w^{[1]} \\ w^{[2]} \end{bmatrix} = \begin{bmatrix} y^{[1]} \\ y^{[2]} \end{bmatrix} \end{matrix}\]
with  $\sigma_L \gg \sigma_S > 0$ . We run gradient descent with a single learning rate  $\eta$  on the squared loss.

\subsection*{5.1 Mode-wise behavior}
\begin{enumerate}
  \item[(a)] Argue that each coordinate  $w^{[1]}$ ,  $w^{[2]}$  behaves like an independent scalar gradient descent on a quadratic with  $\sigma$  equal to  $\sigma_L$  and  $\sigma_S$ , respectively.
  \item[(b)] Write the error recurrences  $e_t^{(1)}$  and  $e_t^{(2)}$  and the corresponding ratios  $r_L(\eta)$  and  $r_S(\eta)$ .
\end{enumerate}

\subsection*{5.2 Stability and limiting factor}
\begin{enumerate}
  \item[(a)] What  $\eta$  range must hold to keep both dimensions stable?
  \item[(b)] Which  $\sigma$  ( $\sigma_L$  or  $\sigma_S$ ) limits how large  $\eta$  can be while still converging?

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\end{enumerate}

\subsection*{5.3 Relative convergence speeds}
\begin{enumerate}
  \item[(a)] For a fixed  $\eta$  in the stable range, which component (associated with  $\sigma_L$  or  $\sigma_S$ ) converges faster?
  \item[(b)] Define a condition number  $\kappa = \sigma_L^2 / \sigma_S^2$ . In one sentence, describe how  $\kappa$  affects the overall convergence rate.
\end{enumerate}

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\section{Gradient Descent and SGD on Simple Models}

You have  $N$  data points  $(x_i, y_i)$ .

\subsection*{6.1 Constant model:  $\hat{y}_i = \theta$ }
We want to minimize

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \theta)^2.$$

\begin{enumerate}
  \item[(a)] Compute  $\frac{\partial L}{\partial \theta}$ .
  \item[(b)] Solve for the optimal  $\theta^*$  in closed form.
  \item[(c)] Write the full-batch gradient descent update  $\theta \leftarrow \theta - \alpha \frac{\partial L}{\partial \theta}$ .
  \item[(d)] Write the SGD update when you sample a single  $i$  uniformly from  $\{1, \dots, N\}$ .
  \item[(e)] In one sentence, explain why each point has the same "vote" in the gradient.
\end{enumerate}

\subsection*{6.2 Linear model:  $\hat{y}_i = \theta x_i$ }
Now

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \theta x_i)^2.$$

\begin{enumerate}
  \item[(a)] Compute  $\frac{\partial L}{\partial \theta}$ .
  \item[(b)] Solve for  $\theta^*$  in closed form.
  \item[(c)] Write the full-batch gradient descent update for  $\theta$ .
  \item[(d)] Write the single-sample SGD update.
  \item[(e)] Do all data points contribute equally in magnitude to the full gradient? Explain in terms of  $x_i$ .
\end{enumerate}

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\section{Noise as Regularization}

For each data point  $\mathbf{x}_i \in \mathbb{R}^n$  we see noisy versions

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i + \mathbf{N}_i, \quad \mathbf{N}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

\text{i.i.d.}
We consider minimizing the expected least-squares loss

$$\min_{\mathbf{w}} \mathbb{E}[\|\tilde{\mathbf{X}} \mathbf{w} - \mathbf{y}\|_2^2].$$


\subsection*{7.1 One-dimensional sanity check}
Let  $x, y, w \in \mathbb{R}$ ,  $\tilde{x} = x + N$ ,  $N \sim \mathcal{N}(0, \sigma^2)$ .
\begin{enumerate}
  \item[(a)] Compute  $\mathbb{E}[\|\tilde{x} w - y\|^2]$  explicitly by expanding and using  $\mathbb{E}[N] = 0$  and  $\mathbb{E}[N^2] = \sigma^2$ .
  \item[(b)] Show that this can be written as  $(x w - y)^2 + (\text{something}) \cdot w^2$ . Identify that "something" in terms of  $x$  and  $\sigma^2$ .
\end{enumerate}

\subsection*{7.2 Matrix case}
Let  $X$  be the  $m \times n$  matrix of true inputs and  $\tilde{X}$  the random noisy matrix with  $i$ -th row  $\tilde{\mathbf{x}}_i^{\text{top}}$ .

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\begin{enumerate}
  \item[(a)] Write  $\|\tilde{X} \mathbf{w} - \mathbf{y}\|_2^2$  as an inner product  $(\tilde{X} \mathbf{w} - \mathbf{y})^\top (\tilde{X} \mathbf{w} - \mathbf{y})$ .
  \item[(b)] Take expectation and split into  $\mathbb{E}[\tilde{X}^\top \tilde{X}]$  terms and cross terms, using that  $\mathbb{E}[\mathbf{w}_i] = \mathbf{0}$  and  $\text{cov}(\mathbf{w}_i) = \sigma^2 \mathbf{I}$ .
  \item[(c)] Show that  $\mathbb{E}[\|\tilde{X} \mathbf{w} - \mathbf{y}\|_2^2]$  can be written as  $\mathbf{X}^\top \mathbf{X} \mathbf{w} - \mathbf{y}^\top \mathbf{X} \mathbf{w} + \lambda \|\mathbf{w}\|_2^2$  for some  $\lambda$ . Identify  $\lambda$  in terms of  $\sigma^2$  and possibly  $\mathbf{X}$ .
\end{enumerate}

\subsection*{7.3 Concept}
\begin{enumerate}
  \item[(a)] In words, why does adding isotropic Gaussian noise to inputs act like an  $\ell_2$  regularizer?
  \item[(b)] If  $\sigma^2$  increases, what happens to the effective  $\lambda$ ?
\end{enumerate}

\bigskip

%-----
\section{General Tikhonov / Weighted Least Squares}

Consider the optimization problem

$$\min_{\mathbf{x}} \|\mathbf{W}_1 (\mathbf{A} \mathbf{x} - \mathbf{b})\|_2^2 + \|\mathbf{W}_2 (\mathbf{x} - \mathbf{c})\|_2^2,$$

where  $\mathbf{W}_1, \mathbf{W}_2, \mathbf{A}$  are matrices and  $\mathbf{x}, \mathbf{b}, \mathbf{c}$  are vectors.

\subsection*{8.1 1D toy version}
Let  $a, b, c, \alpha > 0$  be scalars and solve

$$\min_x \|(a x - b)^2 + \alpha (x - c)^2.$$

\begin{enumerate}
  \item[(a)] Expand the expression and take the derivative with respect to  $x$ .
  \item[(b)] Set the derivative to zero and solve for  $x^*$ .
  \item[(c)] Interpret  $x^*$  as a weighted average of  $b/a$  and  $c$ . What are the weights?
\end{enumerate}

\subsection*{8.2 Matrix expansion}
Back to the matrix problem:
\begin{enumerate}
  \item[(a)] Write the objective as

$$\|\mathbf{A} \mathbf{x} - \mathbf{b}\|_{\mathbf{W}_1}^2 + \|\mathbf{x} - \mathbf{c}\|_{\mathbf{W}_2}^2 = (\mathbf{A} \mathbf{x} - \mathbf{b})^\top \mathbf{W}_1 (\mathbf{A} \mathbf{x} - \mathbf{b}) + (\mathbf{x} - \mathbf{c})^\top \mathbf{W}_2 (\mathbf{x} - \mathbf{c}).$$

  \item[(b)] Take the gradient with respect to  $\mathbf{x}$ , set it to zero, and solve for  $\mathbf{x}^*$  in terms of  $\mathbf{A}, \mathbf{b}, \mathbf{c}, \mathbf{W}_1$ , and  $\mathbf{W}_2$ .
\end{enumerate}

\subsection*{8.3 Stacking trick}
\begin{enumerate}
  \item[(a)] Construct a matrix  $\mathbf{C}$  and vector  $\mathbf{d}$  such that

$$\|\mathbf{W}_1 (\mathbf{A} \mathbf{x} - \mathbf{b})\|_2^2 + \|\mathbf{W}_2 (\mathbf{x} - \mathbf{c})\|_2^2 = \|\mathbf{C} \mathbf{x} - \mathbf{d}\|_2^2.$$

    Explicitly write  $\mathbf{C}$  and  $\mathbf{d}$  as a vertical concatenation.
  \item[(b)] Using the OLS solution  $\mathbf{x}^* = (\mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top \mathbf{d}$ , verify that it matches the expression from 8.2.
  \item[(c)] Choose  $\mathbf{W}_1, \mathbf{W}_2$ , and  $\mathbf{c}$  so that this reduces exactly to standard ridge regression

$$\min_{\mathbf{x}} \|\mathbf{A} \mathbf{x} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{x}\|_2^2.$$

\end{enumerate}

\bigskip

%-----
\section{Gaussian MAP and Ridge Regression}

Let  $\mathbf{W} \in \mathbb{R}^d$  and  $\mathbf{Y} \in \mathbb{R}^n$  be random vectors with

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\begin{align*}
\text{Prior: } & W \sim \text{mathcal{N}}(0, I_d), \quad \\
\text{Noise: } & N \sim \text{mathcal{N}}(0, I_n), \quad \text{quad } N \perp W, \quad \\
\text{Observation model: } & Y = X W + \sqrt{\lambda} N.
\end{align*}

\subsection*{9.1 Covariances}
\begin{enumerate}
\item(a) Compute  $\Sigma_{WW} = \mathbb{E}[W W^\top]$ .
\item(b) Compute  $\Sigma_{YY} = \mathbb{E}[Y Y^\top]$  by expanding  $Y = X W + \sqrt{\lambda} N$  and using independence.
\item(c) Compute  $\Sigma_{WY} = \mathbb{E}[W Y^\top]$ .
\end{enumerate}

\subsection*{9.2 Conditional expectation formula}

For jointly zero-mean Gaussian  $(W, Y)$ :

$$\mathbb{E}[W \mid Y = y] = \Sigma_{WY} \Sigma_{YY}^{-1} y.$$

\begin{enumerate}
\item(a) Plug your matrices from 9.1 into this formula and simplify  $\mathbb{E}[W \mid Y = y]$  into an expression involving  $X$ ,  $\lambda$ , and  $y$  only.
\item(b) Show that this expression matches one of the standard ridge forms

$$\hat{w} = (X^\top X + \lambda I)^{-1} X^\top y$$


$$\text{or}$$


$$\hat{w} = X^\top (X X^\top + \lambda I)^{-1} y.$$

Indicate clearly which one you get and how to see it equals the other.
\end{enumerate}

\subsection*{9.3 Interpretation}
\begin{enumerate}
\item(a) Explain why the MAP estimate of  $W$  given  $Y = y$  is the same as the ridge regression solution.
\item(b) In words, how does  $\lambda$  control how much you "trust" the prior versus the data?
\end{enumerate}

\end{document}

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2 Solutions LaTeX (eecs182_hw1_warmup_solutions.tex)

```

\documentclass[11pt]{article}
\usepackage[margin=1in]{geometry}
\usepackage{amsmath,amssymb,amsfonts}
\usepackage{bm}

\title{EECS 182 -- Homework 1 Warm-Up Worksheet\ Solutions}
\author{}
\date{}

\begin{document}
\maketitle

\section{Vector and Matrix Calculus Refresher}

Let  $x, c \in \mathbb{R}^n$  and  $A \in \mathbb{R}^{n \times n}$ .

\subsection*{1.1 quad  $f(x) = x^\top c$ }
\begin{enumerate}
\item(a)  $\frac{\partial f}{\partial x}$  is a row vector of length  $n$ .
\item(b) Write  $f(x) = \sum_{i=1}^n x_i c_i$ . Then

$$\frac{\partial f}{\partial x_i} = c_i$$


$$\quad \rightarrow \quad$$

\end{enumerate}

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\frac{\partial f}{\partial \bm{x}} = \bm{c}^{\top}.
\]
\end{enumerate}

\subsection*{1.2 \quad $f(\bm{x}) = \bm{x}^{\top} \bm{x}$}
\begin{enumerate}
\item[(a)] $f(\bm{x}) = \sum_{i=1}^n x_i^2$.
\item[(b)] Then
\[\frac{\partial f}{\partial x_i} = 2 x_i\]
\quad \(\Rightarrow\) \quad
\[\frac{\partial f}{\partial \bm{x}} = 2 \bm{x}^{\top}.\]
\]
\end{enumerate}

\subsection*{1.3 \quad $\bm{f}(\bm{x}) = A \bm{x}$}
\begin{enumerate}
\item[(a)] $\frac{\partial \bm{f}}{\partial \bm{x}}$ is an $n \times n$ Jacobian matrix.
\item[(b)] Since $\bm{f}(\bm{x}) = A \bm{x}$ is linear, the Jacobian is just $A$.
\end{enumerate}

\subsection*{1.4 \quad $f(\bm{x}) = \bm{x}^{\top} A \bm{x}$}
\begin{enumerate}
\item[(a)] Writing $A = (a_{ij})$,
\[\bm{f}(\bm{x}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j.\]
\item[(b)] Fix $i$ and differentiate:
\[\frac{\partial f}{\partial x_i} = \sum_j a_{ij} x_j + \sum_j a_{ji} x_j = (A \bm{x})_i + (A^{\top} \bm{x})_i.\]
\item[(c)] Stacking over $i$,
\[\frac{\partial f}{\partial \bm{x}} = \bm{x}^{\top} (A + A^{\top}).\]
\item[(d)] If $A$ is symmetric ($A = A^{\top}$), then
\[\frac{\partial f}{\partial \bm{x}} = 2 \bm{x}^{\top} A.\]
\end{enumerate}

\section{SVD, Least Squares, and Minimum-Norm Solutions}

Let $X \in \mathbb{R}^{m \times n}$, $\bm{y} \in \mathbb{R}^m$, and $X = U \Sigma V^{\top}$.

\subsection*{2.1 Overdetermined case ($m > n$, full column rank)}
\begin{enumerate}
\item[(a)] The normal equations give
\[\begin{aligned} X^{\top} X \bm{w}^{\star} &= X^{\top} \bm{y} \\ \quad \Rightarrow \quad \bm{w}^{\star} &= (X^{\top} X)^{-1} X^{\top} \bm{y}. \end{aligned}\]
\item[(b)] Using $X = U \Sigma V^{\top}$,
\[\begin{aligned} X^{\top} X &= V \Sigma^{\top} U^{\top} U \Sigma V^{\top} = V (\Sigma^{\top} \Sigma) V^{\top}. \end{aligned}\]
Hence
\[\begin{aligned} (X^{\top} X)^{-1} &= V (\Sigma^{\top} \Sigma)^{-1} V^{\top}. \end{aligned}\]
Therefore
\begin{align*} \bm{w}^{\star} &= (X^{\top} X)^{-1} X^{\top} \bm{y} \\ &= V (\Sigma^{\top} \Sigma)^{-1} V^{\top} (V \Sigma^{\top} U^{\top}) \bm{y} \\ &= V (\Sigma^{\top} \Sigma)^{-1} \Sigma^{\top} U^{\top} \bm{y}. \end{align*}
Since $\Sigma^{\top} \Sigma$ is diagonal with entries $\sigma_i^2$, $(\Sigma^{\top} \Sigma)^{-1}$ is diagonal with entries $1/\sigma_i$. This is precisely $\Sigma^{+}$, so

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\[\bm{w}^{\star} = V \Sigma^{+} U^{\top} \bm{y}.\]
\end{enumerate}

\subsection*{2.2 Underdetermined case ( $m < n$ , full row rank)}
\begin{enumerate}
\item[(a)] We solve
\[\min_{\|\bm{w}\|_2^2} \text{subject to} \quad X \bm{w} = \bm{y}.\]
\item[(b)] Form the Lagrangian
\[\mathcal{L}(\bm{w}, \bm{\lambda}) = \|\bm{w}\|_2^2 + \bm{\lambda}^{\top} (X \bm{w} - \bm{y}).\]
Setting the derivative with respect to  $\bm{w}$  to zero:
\[\begin{aligned} 2 \bm{w} + X^{\top} \bm{\lambda} &= 0 \\ \Rightarrow \bm{w} &= -\frac{1}{2} X^{\top} \bm{\lambda}. \end{aligned}\]
Substitute into  $X \bm{w} = \bm{y}$ :
\[\begin{aligned} X \left(-\frac{1}{2} X^{\top} \bm{\lambda}\right) &= \bm{y} \\ \Rightarrow -\frac{1}{2} X X^{\top} \bm{\lambda} &= \bm{y}. \end{aligned}\]
Thus  $\bm{\lambda} = -2 (X X^{\top})^{-1} \bm{y}$ , and
\[\bm{w}_{\min} = X^{\top} (X X^{\top})^{-1} \bm{y}.\]
\item[(c)] Using the SVD,  $X X^{\top} = U \Sigma \Sigma^{\top} U^{\top}$ , so
\[(X X^{\top})^{-1} = U (\Sigma \Sigma^{\top})^{-1} U^{\top}.\]
Then
\begin{align*} \bm{w}_{\min} &= X^{\top} (X X^{\top})^{-1} \bm{y} \\ &= V \Sigma^{\top} U^{\top} U (\Sigma \Sigma^{\top})^{-1} U^{\top} \bm{y} \\ &= V \Sigma^{\top} (\Sigma \Sigma^{\top})^{-1} U^{\top} \bm{y}. \end{align*}
The diagonal matrix  $\Sigma \Sigma^{\top}$  has entries  $1/\sigma_i$ , so it is  $\Sigma^{+}$ . Thus
\[\bm{w}_{\min} = V \Sigma^{+} U^{\top} \bm{y}.\]
\end{enumerate}

\subsection*{2.3 Concept check}
\begin{enumerate}
\item[(a)] A least-squares solution minimizes the residual norm  $\|X \bm{w} - \bm{y}\|_2$ . A minimum-norm solution is the solution among all exact fits  $X \bm{w} = \bm{y}$  that has the smallest  $\|\bm{w}\|_2$ .
\item[(b)] With full column rank ( $m > n$ ), the least-squares solution is unique. With full row rank ( $m < n$ ), the minimum-norm solution is unique.
\end{enumerate}

\section{Ridge Regression Basics}

Let  $X \in \mathbb{R}^{n \times d}$ ,  $\bm{y} \in \mathbb{R}^n$ ,  $\lambda > 0$ , and
\[\mathcal{L}(\bm{w}) = \|\bm{y} - X \bm{w}\|_2^2 + \lambda \|\bm{w}\|_2^2.\]

\subsection*{3.1 Closed-form solution}
\begin{enumerate}
\item[(a)] Expand:
\[\mathcal{L}(\bm{w}) = (\bm{y} - X \bm{w})^{\top} (\bm{y} - X \bm{w}) + \lambda \bm{w}^{\top} \bm{w}.\]
The gradient is
\[\begin{aligned} \frac{\partial \mathcal{L}}{\partial \bm{w}} &= -2 X^{\top} (\bm{y} - X \bm{w}) + 2 \lambda \bm{w}. \end{aligned}\]

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\]
\item[(b)] Setting this to zero:
\[
-2 X^{\top} (\bm{y} - X \bm{w}) + 2 \lambda \bm{w} = 0
\]
gives
\[
X^{\top} X \bm{w} - X^{\top} \bm{y} + \lambda \bm{w} = 0
\quad \Rightarrow \quad
(X^{\top} X + \lambda I) \bm{w} = X^{\top} \bm{y}.
\]
So
\[
\bm{w}_{\text{ridge}} = (X^{\top} X + \lambda I)^{-1} X^{\top} \bm{y}.
\]
\end{enumerate}

\subsection*{3.2 SVD perspective and shrinkage}

Let  $X = U \Sigma V^{\top}$ .
\begin{enumerate}
\item[(a)] Then  $X^{\top} X = V \Sigma^{\top} \Sigma V^{\top}$ , so
\[
\bm{w}_{\text{ridge}} = V (\Sigma^{\top} \Sigma + \lambda I)^{-1} \Sigma^{\top} U^{\top} \bm{y}.
\]
The only non-identity diagonal matrix is  $(\Sigma^{\top} \Sigma + \lambda I)^{-1}$ 
 $\Sigma^{\top}$ , which acts diagonally on singular directions.
\item[(b)] Suppose  $U^{\top} \bm{y}$  has components  $\alpha_i$  along singular vectors. In the
 $V$ -basis, the coefficient along direction  $i$  is
\[
\frac{\sigma_i}{\sigma_i^2 + \lambda} \alpha_i.
\]
Equivalently, relative to the OLS coefficient (which scales like  $\alpha_i / \sigma_i$ ),
ridge multiplies by
\[
\frac{\sigma_i^2}{\sigma_i^2 + \lambda}.
\]
\item[(c)] If  $\sigma_i$  is very small,  $\frac{\sigma_i^2}{\sigma_i^2 + \lambda} \approx 0$ ,
so that direction is heavily shrunk. If  $\sigma_i$  is very large, the factor is close to 1, so
that direction is only mildly shrunk.
\end{enumerate}

\subsection*{3.3 Interpretation}
\begin{enumerate}
\item[(a)] Ridge is called "shrinkage" because it shrinks coefficients toward zero,
especially along directions that  $X$  does not constrain well (those with small  $\sigma_i$ ).
\item[(b)] As  $\lambda \rightarrow 0$ ,  $\bm{w}_{\text{ridge}} \rightarrow (X^{\top} X)^{-1} X^{\top} \bm{y}$ ,
the ordinary least-squares solution (assuming  $X^{\top} X$  is invertible).
\item[(c)] As  $\lambda \rightarrow \infty$ , the penalty dominates and  $\bm{w}_{\text{ridge}} \rightarrow \bm{0}$ .
\end{enumerate}

\section{Scalar Gradient Descent Stability}

Consider  $L(w) = (y - \sigma w)^2$  with  $\sigma > 0$  and gradient descent
\[
w_{t+1} = w_t - \eta \frac{dL}{dw} \Big|_{w_t}.
\]

\subsection*{4.1 Derive the update}
\begin{enumerate}
\item[(a)] Expand:
\[
L(w) = (y - \sigma w)^2 = (\sigma w - y)^2.
\]
Then
\[
\frac{dL}{dw} = 2 (\sigma w - y) \sigma = 2 \sigma (\sigma w - y).
\]
\item[(b)] The update is
\begin{align*}
w_{t+1} &= w_t - \eta \cdot 2 \sigma (\sigma w_t - y) \\
&= w_t - 2 \eta \sigma^2 w_t + 2 \eta \sigma y
\end{align*}



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&= (1 - 2 \eta \sigma^2) w_t + 2 \eta \sigma y.
\end{align*}
\item[(c)] The optimal  $w^{\star}$  solves  $\frac{dL}{dw} = 0$ , i.e.  $\sigma w^{\star} - y = 0$ ,
so
\begin{aligned}
w^{\star} &= \frac{y}{\sigma}.
\end{aligned}
\end{enumerate}

\subsection*{4.2 Error dynamics}
Let  $e_t = w_t - w^{\star}$ .
\begin{enumerate}
\item[(a)] Write
\begin{aligned}
w_t &= e_t + w^{\star}.
\end{aligned}
Then
\begin{align*}
w_{t+1} &= (1 - 2 \eta \sigma^2) (e_t + w^{\star}) + 2 \eta \sigma y.
\end{align*}
Using  $w^{\star} = y/\sigma$ ,
\begin{aligned}
&(1 - 2 \eta \sigma^2) w^{\star} + 2 \eta \sigma y \\
&= (1 - 2 \eta \sigma^2) \frac{y}{\sigma} + 2 \eta \sigma y \\
&= \frac{y}{\sigma} = w^{\star}.
\end{aligned}
\end{aligned}
Hence
\begin{aligned}
w_{t+1} - w^{\star} &= (1 - 2 \eta \sigma^2) e_t.
\end{aligned}
\item[(b)] So
\begin{aligned}
e_{t+1} &= (1 - 2 \eta \sigma^2) e_t,
\end{aligned}
and by iteration,
\begin{aligned}
e_t &= (1 - 2 \eta \sigma^2)^t e_0.
\end{aligned}
The common ratio is
\begin{aligned}
r(\eta, \sigma) &= 1 - 2 \eta \sigma^2.
\end{aligned}
\end{enumerate}

\subsection*{4.3 Stability region}
\begin{enumerate}
\item[(a)] We need  $|r| < 1$  for convergence, i.e.
\begin{aligned}
|1 - 2 \eta \sigma^2| &< 1.
\end{aligned}
\item[(b)] This inequality is equivalent to
\begin{aligned}
-1 &< 1 - 2 \eta \sigma^2 < 1 \\
&\quad \Rightarrow \\
0 &< 2 \eta \sigma^2 < 2 \\
&\quad \Rightarrow \\
0 &< \eta < \frac{1}{\sigma^2}.
\end{aligned}
\end{enumerate}

\subsection*{4.4 Convergence speed}
\begin{enumerate}
\item[(a)] If  $|e_t| \leq \alpha^t |e_0|$  with  $\alpha \in (0, 1)$ , then to ensure  $|e_t| \leq \epsilon$  we need
\begin{aligned}
&\alpha^t \leq \epsilon \\
&\quad \Rightarrow \\
t &\geq \frac{\log(\epsilon)}{\log(\alpha)}.
\end{aligned}
(Note  $\log(\alpha) < 0$ .)
\item[(b)] Here  $\alpha = |1 - 2 \eta \sigma^2|$ , so
\begin{aligned}
t &\geq \frac{\log(\epsilon)}{\log |1 - 2 \eta \sigma^2|}.
\end{aligned}
\end{enumerate}

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\end{enumerate}

\section{Two-Dimensional Diagonal System and Condition Number}

The system is
\begin{bmatrix}
\sigma_L & 0 \\
0 & \sigma_S
\end{bmatrix}
\begin{bmatrix}
w^{[1]} \\
w^{[2]}
\end{bmatrix}
=
\begin{bmatrix}
y^{[1]} \\
y^{[2]}
\end{bmatrix},
\quad \sigma_L \gg \sigma_S > 0.
\end{bmatrix}

\subsection*{5.1 Mode-wise behavior}
\begin{enumerate}
\item(a) The squared loss decomposes as a sum of two independent 1D losses, one involving  $(\sigma_L, w^{[1]}, y^{[1]})$  and one involving  $(\sigma_S, w^{[2]}, y^{[2]})$ . Thus each coordinate follows the scalar gradient descent dynamics from Section-4 with its own  $\sigma$ .
\item(b) For  $k \in \{1, 2\}$  with  $\sigma_k \in \{\sigma_L, \sigma_S\}$ , the error satisfies
\begin{bmatrix}
e_{t+1}^{(k)} = (1 - 2 \eta \sigma_k^2) e_t^{(k)}.
\end{bmatrix}
Hence
\begin{bmatrix}
r_L(\eta) = 1 - 2 \eta \sigma_L^2, \\
r_S(\eta) = 1 - 2 \eta \sigma_S^2.
\end{bmatrix}
\end{enumerate}

\subsection*{5.2 Stability and limiting factor}
\begin{enumerate}
\item(a) We require  $|1 - 2 \eta \sigma_L^2| < 1$  and  $|1 - 2 \eta \sigma_S^2| < 1$ . Since  $\sigma_L > \sigma_S$ , the first condition is stricter. Hence we must choose
\begin{bmatrix}
0 < \eta < \frac{1}{\sigma_L^2}.
\end{bmatrix}
\item(b) The larger singular value  $\sigma_L$  limits how large  $\eta$  can be while still ensuring stability for both coordinates.
\end{enumerate}

\subsection*{5.3 Relative convergence speeds}
\begin{enumerate}
\item(a) For a fixed  $\eta$  in the stable range, the factor  $|1 - 2 \eta \sigma^2|$  is smaller when  $\sigma$  is larger (up to the stability limit). Thus the component associated with  $\sigma_L$  converges faster than the one associated with  $\sigma_S$ .
\item(b) The condition number  $\kappa = \sigma_L^2 / \sigma_S^2$  measures how stretched the problem is. Large  $\kappa$  forces  $\eta$  to be small (to keep the large- $\sigma$  mode stable), which makes the small- $\sigma$  mode converge very slowly, slowing overall convergence.
\end{enumerate}

\section{Gradient Descent and SGD on Simple Models}

\subsection*{6.1 Constant model:  $\hat{y}_i = \theta$ }

We have
\begin{bmatrix}
L(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \theta)^2.
\end{bmatrix}
\begin{enumerate}
\item(a) Differentiate:
\begin{bmatrix}
\frac{\partial L}{\partial \theta} \\
= \frac{1}{N} \sum_i 2 (y_i - \theta)(-1) \\
= \frac{2}{N} \sum_i (\theta - y_i).
\end{bmatrix}
\end{enumerate}

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\]
\item[(b)] Set to zero:
\[
\frac{2}{N} \sum_i (\theta - y_i) = 0
\quad \Rightarrow \quad
\theta \sum_i 1 = \sum_i y_i
\quad \Rightarrow \quad
\theta^* = \frac{1}{N} \sum_{i=1}^N y_i.
\]
\item[(c)] Full-batch GD:
\[
\theta \rightarrow \theta - \alpha \cdot \frac{2}{N} \sum_{i=1}^N (\theta - y_i).
\]
\item[(d)] Single-sample SGD (for a sampled  $i$ ):
\[
\theta \rightarrow \theta - \alpha \cdot 2 (\theta - y_i).
\]
\item[(e)] Each point contributes a term of the same functional form  $(\theta - y_i)$ , with
no  $x_i$  scaling, so every data point has an equal "vote" in the gradient.
\end{enumerate}

\subsection*{6.2 Linear model:  $\hat{y}_i = \theta x_i$ }

Now
\[
L(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \theta x_i)^2.
\]
\begin{enumerate}
\item[(a)] Differentiate:
\[
\frac{\partial L}{\partial \theta} = \frac{1}{N} \sum_i 2 (y_i - \theta x_i) (-x_i) = -\frac{2}{N} \sum_i (x_i y_i - x_i^2 \theta).
\]
\item[(b)] Set to zero:
\[
\frac{2}{N} \sum_i (\theta x_i^2 - x_i y_i) = 0
\quad \Rightarrow \quad
\theta \sum_i x_i^2 = \sum_i x_i y_i,
\]
so
\[
\theta^* = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.
\]
\item[(c)] Full-batch GD update:
\[
\theta \rightarrow \theta - \alpha \cdot \frac{2}{N} \sum_{i=1}^N (\theta x_i - y_i) x_i.
\]
\item[(d)] Single-sample SGD (for a sampled  $i$ ):
\[
\theta \rightarrow \theta - \alpha \cdot 2 (\theta x_i - y_i) x_i.
\]
\item[(e)] No: points do not contribute equally in magnitude. Their contributions are scaled
by  $|x_i|$  (or  $x_i^2$  in the sum), so points with larger  $|x_i|$  have larger influence on the
gradient.
\end{enumerate}

\section{Noise as Regularization}

\subsection*{7.1 One-dimensional sanity check}

Let  $\tilde{x} = x + N$ ,  $N \sim \mathcal{N}(0, \sigma^2)$ , and consider  $(\tilde{x} w - y)^2$ .
\begin{enumerate}
\item[(a)] Expand:
\begin{align*}
&(\tilde{x} w - y)^2 \\
&= ((x + N) w - y)^2 \\
&= (x w - y + N w)^2 \\
&= (x w - y)^2 + 2 (x w - y) (N w) + (N w)^2.
\end{align*}
\end{enumerate}
Taking expectation:
\begin{align*}
\mathbb{E}[(x w - y)^2] &= (x w - y)^2 \\
&\quad (\text{no randomness}),
\end{align*}

```

```

and  $\mathbb{E}[N] = 0$  so the cross term vanishes. Also,
\[
\mathbb{E}[(N w)^2] = w^2 \mathbb{E}[N^2] = w^2 \sigma^2.
\]
Thus
\[
\mathbb{E}[(\tilde{x} w - y)^2] = (x w - y)^2 + \sigma^2 w^2.
\]
\item{(b)} The ``something'' multiplying  $w^2$  is  $\sigma^2$ .
\end{enumerate}

\subsection*{7.2 Matrix case}

Let  $X$  be  $m \times n$  with rows  $\mathbf{x}_i^{\text{top}}$ , and  $\tilde{X} = X + N$  with rows
 $\tilde{\mathbf{x}}_i^{\text{top}} = \mathbf{x}_i^{\text{top}} + \mathbf{N}_i^{\text{top}}$ .
\begin{enumerate}
\item{(a)} We can write
\[
\|\tilde{X} \mathbf{w} - \mathbf{y}\|_2^2 = (\tilde{X} \mathbf{w} - \mathbf{y})^{\text{top}} (\tilde{X} \mathbf{w} - \mathbf{y})^{\text{bottom}}.
\]
\item{(b)} Substitute  $\tilde{X} = X + N$ :
\[
\tilde{X} \mathbf{w} - \mathbf{y} = (X \mathbf{w} - \mathbf{y}) + N \mathbf{w},
\]
so
\begin{align*}
\|\tilde{X} \mathbf{w} - \mathbf{y}\|_2^2 &= \|X \mathbf{w} - \mathbf{y}\|_2^2 + 2 (X \mathbf{w} - \mathbf{y})^{\text{top}} N \mathbf{w} + \|N \mathbf{w}\|_2^2.
\end{align*}
\end{enumerate}
Taking expectation, the middle term vanishes since each  $\mathbf{N}_i$  has mean  $\mathbf{0}$  and is independent of  $(X \mathbf{w} - \mathbf{y})$ . For the last term:
\[
\|N \mathbf{w}\|_2^2 = \sum_{i=1}^m (\mathbf{N}_i^{\text{top}} \mathbf{w})^2.
\]
Each  $\mathbf{N}_i$  has covariance  $\sigma^2 I$ , so
\[
\mathbb{E}[(\mathbf{N}_i^{\text{top}} \mathbf{w})^2] = \mathbf{w}^{\text{top}} \mathbb{E}[\mathbf{N}_i \mathbf{N}_i^{\text{top}}] \mathbf{w} = \mathbf{w}^{\text{top}} (\sigma^2 I) \mathbf{w} = \sigma^2 \|\mathbf{w}\|_2^2.
\]
Summing over  $i$ :
\[
\mathbb{E}[\|N \mathbf{w}\|_2^2] = m \sigma^2 \|\mathbf{w}\|_2^2.
\]
Thus
\[
\mathbb{E}[\|\tilde{X} \mathbf{w} - \mathbf{y}\|_2^2] = \|X \mathbf{w} - \mathbf{y}\|_2^2 + m \sigma^2 \|\mathbf{w}\|_2^2.
\]
\item{(c)} Hence  $\lambda = m \sigma^2$ .
\end{enumerate}

\subsection*{7.3 Concept}
\begin{enumerate}
\item{(a)} Adding isotropic Gaussian noise to inputs makes the expected loss include an extra term proportional to  $\|\mathbf{w}\|_2^2$ , which is exactly  $\ell_2$  regularization. Intuitively, large weights make predictions very sensitive to input noise, and the optimization penalizes such sensitivity.
\item{(b)} As  $\sigma^2$  increases, the effective  $\lambda$  increases, so regularization becomes stronger and the solution shrinks more toward zero.
\end{enumerate}

\section{General Tikhonov / Weighted Least Squares}

\subsection*{8.1 1D toy version}

Consider
\[
f(x) = (a x - b)^2 + \alpha (x - c)^2, \quad \text{quad } a, b, c, \alpha > 0.
\]
\begin{enumerate}
\item{(a)} Expand:
\begin{align*}
\end{align*}

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```
f(x)
&= (a^2 x^2 - 2ab x + b^2) + (\alpha x^2 - 2 \alpha c x + \alpha c^2) \\\
&= (a^2 + \alpha)x^2 - 2 (ab + \alpha c) x + (b^2 + \alpha c^2).
\end{align*}
Then
\[
\frac{df}{dx} = 2 (a^2 + \alpha)x - 2 (ab + \alpha c).
\]
\item[(b)] Set derivative to zero:
\[
2 (a^2 + \alpha)x - 2 (ab + \alpha c) = 0
\quad \Rightarrow \quad
x^{\star} = \frac{ab + \alpha c}{a^2 + \alpha}.
\]
\item[(c)] Rewrite:
\[
x^{\star} = \frac{a^2}{a^2 + \alpha} \cdot \frac{b}{a} + \frac{\alpha}{a^2 + \alpha} \cdot c.
\]
So  $x^{\star}$  is a weighted average of  $b/a$  and  $c$  with weights proportional to  $a^2$  and  $\alpha$ .
\end{enumerate}
```

\subsection*{8.2 Matrix expansion}

```
For
\[
\min_{\bm{x}} \|W_1 (A \bm{x} - \bm{b})\|_2^2 + \|W_2 (\bm{x} - \bm{c})\|_2^2,
\]
write
\begin{align*}
& (A \bm{x} - \bm{b})^{\top} W_1^{\top} W_1 (A \bm{x} - \bm{b}) \\
& + (\bm{x} - \bm{c})^{\top} W_2^{\top} W_2 (\bm{x} - \bm{c}).
\end{align*}
Taking the gradient with respect to  $\bm{x}$ :
\begin{align*}
& \frac{\partial}{\partial \bm{x}} \\
& = 2 A^{\top} W_1^{\top} W_1 (A \bm{x} - \bm{b}) + 2 W_2^{\top} W_2 (\bm{x} - \bm{c}).
\end{align*}
Setting to zero:
\[
A^{\top} W_1^{\top} W_1 (A \bm{x} - \bm{b}) + W_2^{\top} W_2 (\bm{x} - \bm{c}) = 0.
\]
Rearranging:
\[
(A^{\top} W_1^{\top} W_1 A + W_2^{\top} W_2) \bm{x} \\
= A^{\top} W_1^{\top} W_1 \bm{b} + W_2^{\top} W_2 \bm{c}.
\]
Thus
\[
\bm{x}^{\star} = (A^{\top} W_1^{\top} W_1 A + W_2^{\top} W_2)^{-1} \\
\big(A^{\top} W_1^{\top} W_1 \bm{b} + W_2^{\top} W_2 \bm{c}\big).
\]
```

\subsection*{8.3 Stacking trick}

```
\begin{enumerate}
\item[(a)] Define
\[
C = \begin{bmatrix} W_1 A & W_2 I \end{bmatrix}, \\
\bm{d} = \begin{bmatrix} W_1 \bm{b} & W_2 \bm{c} \end{bmatrix}.
\]
Then
\[
C \bm{x} - \bm{d} \\
= \\
\begin{bmatrix} W_1 (A \bm{x} - \bm{b}) \\ W_2 (\bm{x} - \bm{c}) \end{bmatrix},
\]
and hence
\[
\|C \bm{x} - \bm{d}\|_2^2 \\
= \|W_1 (A \bm{x} - \bm{b})\|_2^2 + \|W_2 (\bm{x} - \bm{c})\|_2^2.
\]
```

```

\]
\item[(b)] Now
\[
C^{\top} C = A^{\top} W_1^{\top} W_1 A + W_2^{\top} W_2,
\quad
C^{\top} \bm{d} = A^{\top} W_1^{\top} W_1 \bm{b} + W_2^{\top} W_2 \bm{c}.
\]
The OLS solution  $\bm{x}^{\star} = (C^{\top} C)^{-1} C^{\top} \bm{d}$  matches the formula in 8.2.
\item[(c)] For ridge regression set  $W_1 = I$ ,  $W_2 = \sqrt{\lambda} I$ ,  $\bm{c} = \bm{0}$ ,
and rename  $A = X$ ,  $\bm{b} = \bm{y}$ . Then
\[
A^{\top} W_1^{\top} W_1 A + W_2^{\top} W_2
= X^{\top} X + \lambda I,
\]
and
\[
A^{\top} W_1^{\top} W_1 \bm{b} + W_2^{\top} W_2 \bm{c}
= X^{\top} \bm{y} + 0,
\]
so
\[
\bm{x}^{\star} = (X^{\top} X + \lambda I)^{-1} X^{\top} \bm{y},
\]
which is exactly ridge regression.
\end{enumerate}

\section{Gaussian MAP and Ridge Regression}

Let  $W \in \mathbb{R}^d$ ,  $Y \in \mathbb{R}^n$  with
\[
W \sim \mathcal{N}(0, I_d), \quad
N \sim \mathcal{N}(0, I_n), \quad
Y = X W + \sqrt{\lambda} N,
\]
\end{equation}

\subsection{9.1 Covariances}
\begin{enumerate}
\item[(a)] By definition,
\[
\Sigma_{WW} = \mathbb{E}[W W^{\top}] = I_d.
\]
\item[(b)] Expand
\[
Y = X W + \sqrt{\lambda} N.
\]
Then
\begin{align*}
\Sigma_{YY} &= \mathbb{E}[Y Y^{\top}] \\
&= \mathbb{E}[(X W + \sqrt{\lambda} N)(X W + \sqrt{\lambda} N)^{\top}] \\
&= \mathbb{E}[X W W^{\top} X^{\top}] + \lambda \mathbb{E}[N N^{\top}] \\
&\quad + \mathbb{E}[X W N^{\top} \sqrt{\lambda} + \sqrt{\lambda} N W^{\top} X^{\top}].
\end{align*}
Independence and zero-mean imply the cross terms vanish, so
\[
\Sigma_{YY} = X \mathbb{E}[W W^{\top}] X^{\top} + \lambda I_n = X X^{\top} + \lambda I.
\]
\item[(c)] For  $\Sigma_{WY}$ :
\begin{align*}
\Sigma_{WY} &= \mathbb{E}[W Y^{\top}] \\
&= \mathbb{E}[W (X W + \sqrt{\lambda} N)^{\top}] \\
&= \mathbb{E}[W W^{\top} X^{\top}] + \sqrt{\lambda} \mathbb{E}[W N^{\top}] \\
&= X^{\top}.
\end{align*}
\end{enumerate}

\subsection{9.2 Conditional expectation and ridge form}

For jointly zero-mean Gaussian  $(W, Y)$ ,
\[
\mathbb{E}[W \mid Y = \bm{y}] = \Sigma_{WY} \Sigma_{YY}^{-1} \bm{y}.
\]
Using the results above:

```

```

\mathbb{E}[W \mid Y = \bm{y}] = X^{\top} (X X^{\top} + \lambda I)^{-1} \bm{y}.
\]
This is one standard form of the ridge solution:
\[
\hat{\bm{w}} = X^{\top} (X X^{\top} + \lambda I)^{-1} \bm{y}.
\]
There is also the equivalent form
\[
\hat{\bm{w}} = (X^{\top} X + \lambda I)^{-1} X^{\top} \bm{y},
\]
and the two are related by the matrix identity
\[
X^{\top} (X X^{\top} + \lambda I)^{-1} = (X^{\top} X + \lambda I)^{-1} X^{\top}.
\]

\subsection*{9.3 Interpretation}
\begin{enumerate}
\item[(a)] The posterior  $p(W \mid Y = \bm{y})$  is Gaussian, and its mean (also the MAP estimator) is exactly the ridge solution. Minimizing the negative log posterior is equivalent to minimizing squared error plus an  $\ell_2$  penalty  $\lambda \|\bm{w}\|_2^2$ .
\item[(b)] Larger  $\lambda$  corresponds to stronger prior/greater noise variance in the observation model. As  $\lambda$  increases, the estimator trusts the prior more and shrinks  $\bm{w}$  more strongly toward  $\mathbf{0}$ ; as  $\lambda$  decreases, it trusts the data more and approaches the unregularized least-squares estimate.
\end{enumerate}

\end{document}

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If you want, I can also quickly compile these to PDFs again so you have “official” LaTeX-based versions matching the earlier ones.